Visual Perception and Memory

temporal resolution

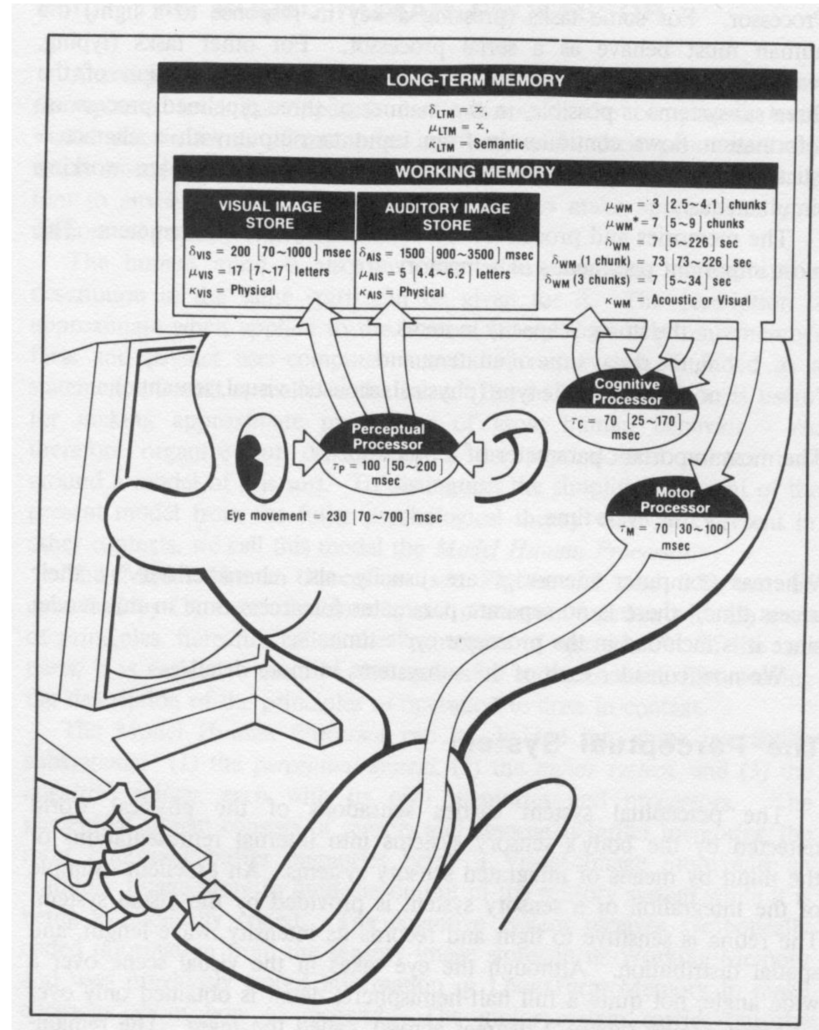
spatial resolution

colour perception

memory

The Human Processor

- Input
  - Senses
- Process
  - Cognition
- Storage
  - Memory
- Output
  - Actions
  - Speech

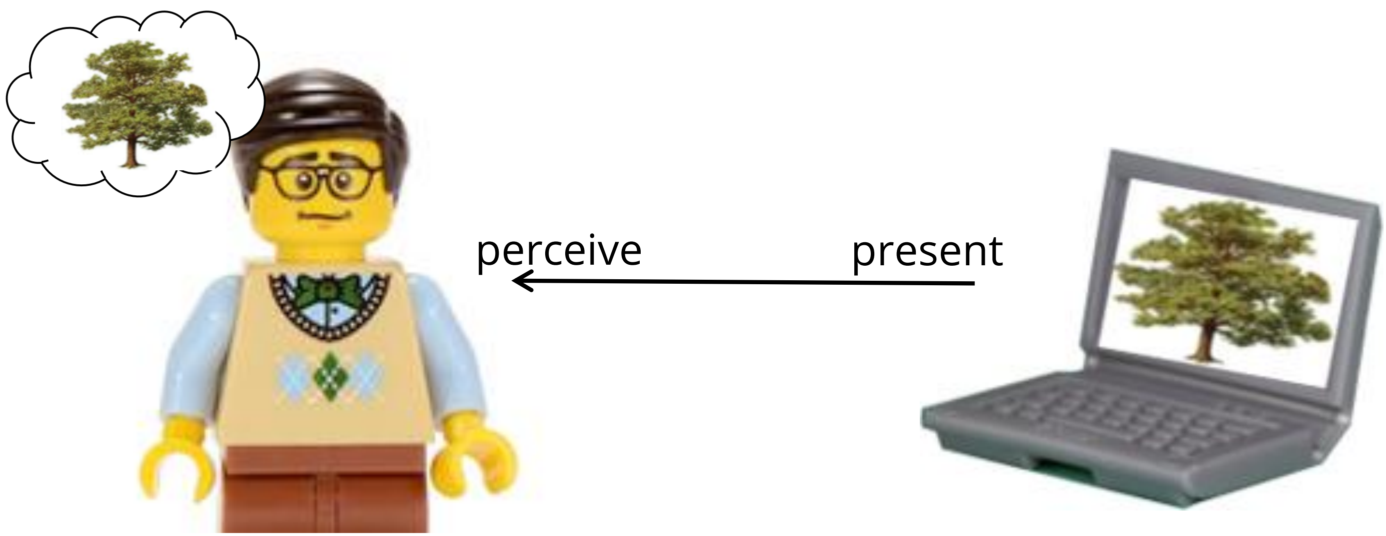


Human Senses and Perception

- Five senses: Processing of sensory information to **see** (visual), **hear** (sound), **taste**, **smell**, or **feel** objects in the world
- Vision: most dominant form of perception
- Understand perceptual phenomena for proper visual designs
  - Need to understand capabilities and limitations
  - What do they find difficult or impossible
- Unfortunately, no simple model to describe humans

Human Vision

- Important to understand human vision
  - Temporal resolution
  - Spatial Resolution
  - Colour Perception



Temporal Resolution: Flicker

- Critical Flicker Frequency (CFF)
  - when perception of intermittent light source changes from flickering to continuous light
  - dependent on brightness of stimulus, wavelength, others …

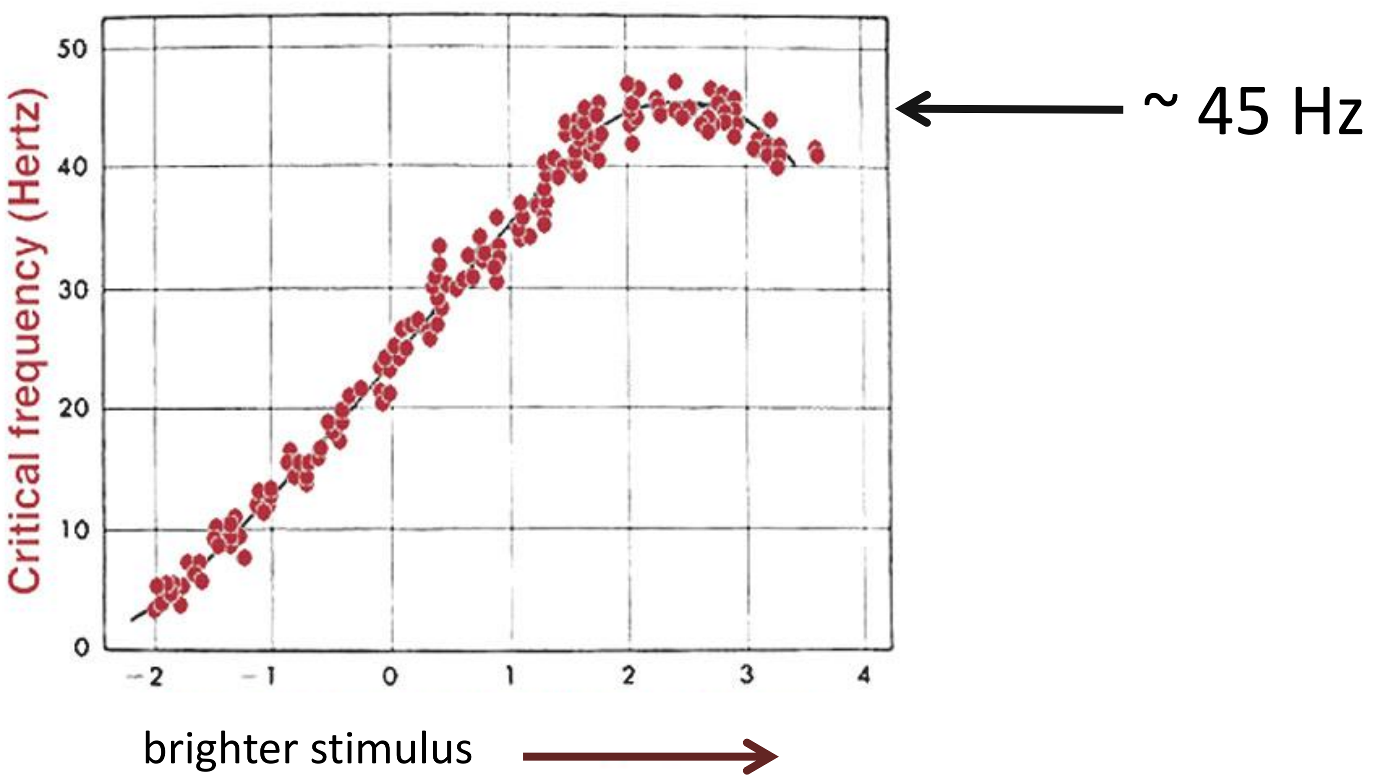


Image: http://webvision.med.utah.edu

Temporal Resolution: Flicker into Motion

- CFF can also create perception of continuous motion
  - motion blur
  - 24 FPS film, 60 FPS NTSC video, HFR video 120 FPS

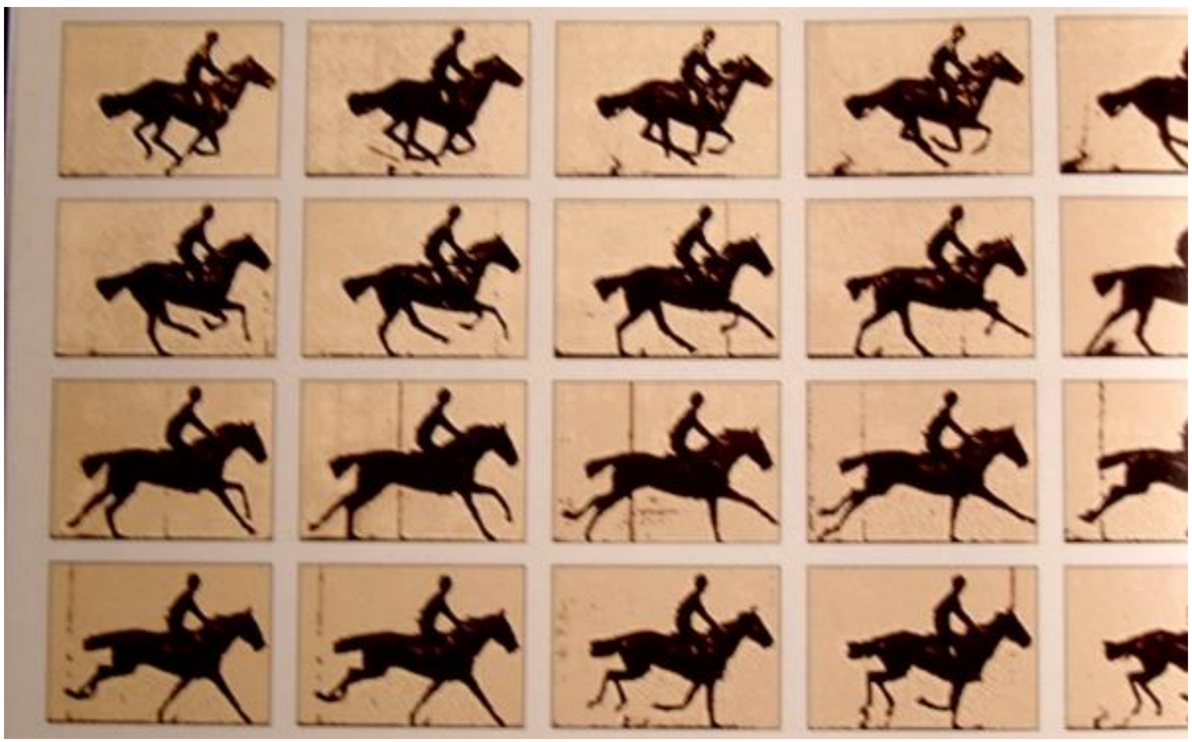


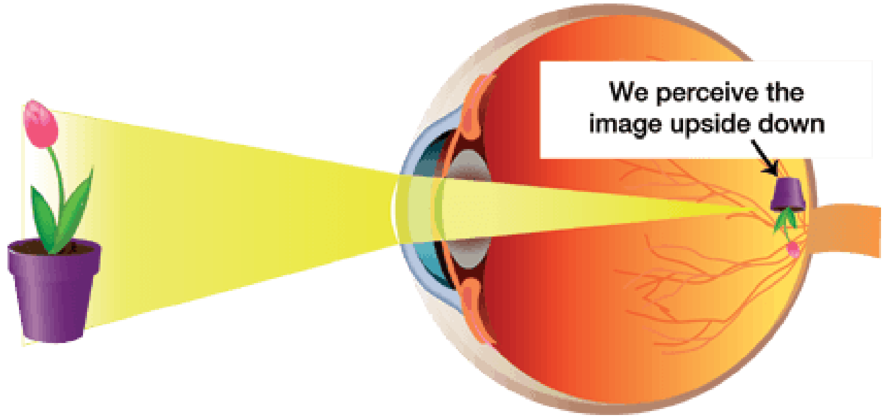
Image: http://webvision.med.utah.edu

Zoetrope, mechanical example of CFF

- https://youtu.be/-hE_fA9M580?t=5s

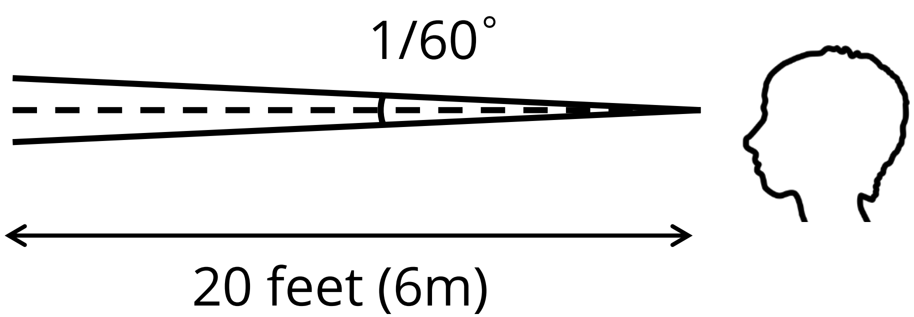
The Eye

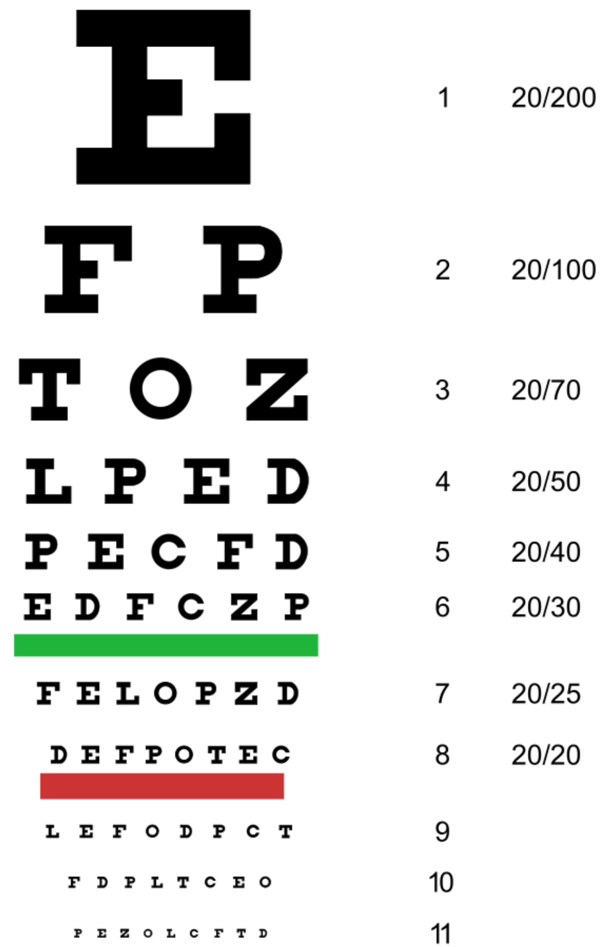
- The eye is a mechanism for receiving light and transforming it into electrical energy
- Light
  - Reflects from objects (e.g. paper)
  - Or is produced from a light source (e.g. a display)
- Images are focused upside-down on retina
- ganglion cells (brain!) detect pattern and movement



Spatial Resolution: Visual Acuity

- spatial resolution of visual processing system
  - 20/20 (6/6) vision: separate lines 1 arc minute (1/60°) apart at 20 feet (6 m)





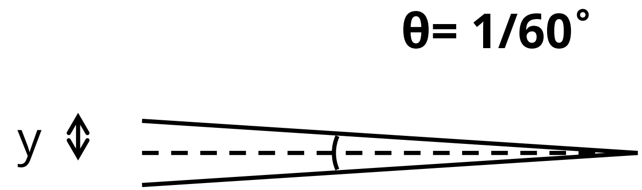
20/20 → Normal vision

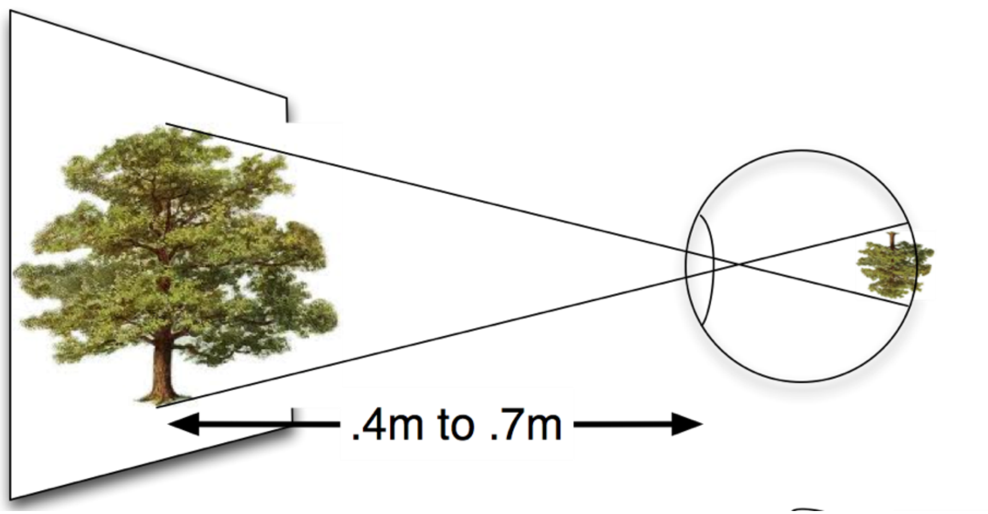
20/15 → A test subject sees at 20 feet that a person with 20/20 vision can see at 15 feet.

20/40 → A test subject sees at 20 feet what a person with normal vision sees at 40 feet

Spatial Resolution Implications

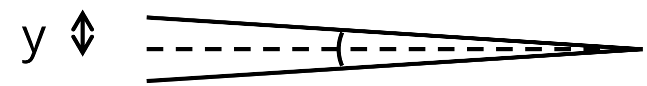
- Best pixel density for displays?
  - density is ppcm (pixels per cm)

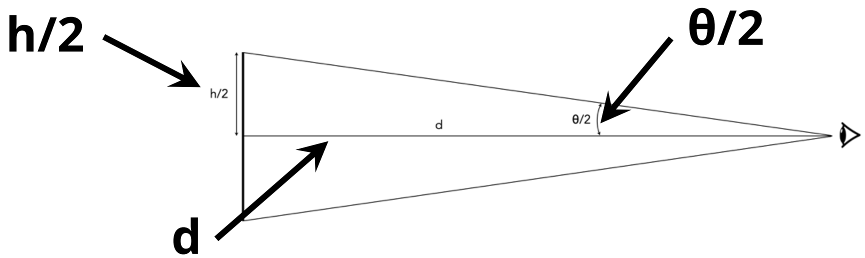




Spatial Resolution Implications

$\theta = 1/60^\circ$





$\tan(\theta/2) = (h/2)/d$

$h/2 = d \tan(\theta/2)$

$h = 2d \times \tan(\theta/2)$

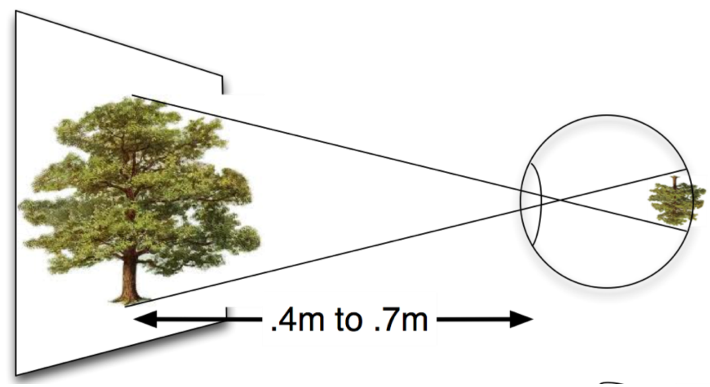
1 arcminute, i.e., $1/60^\circ$, is equivalent to 0.000290888 radians

$h = 2d \times \tan(0.000290888/2)$

$= 2d \times 0.000145444$

Spatial Resolution Implications

- Best pixel density for displays?
  - density is ppcm (pixels per cm)

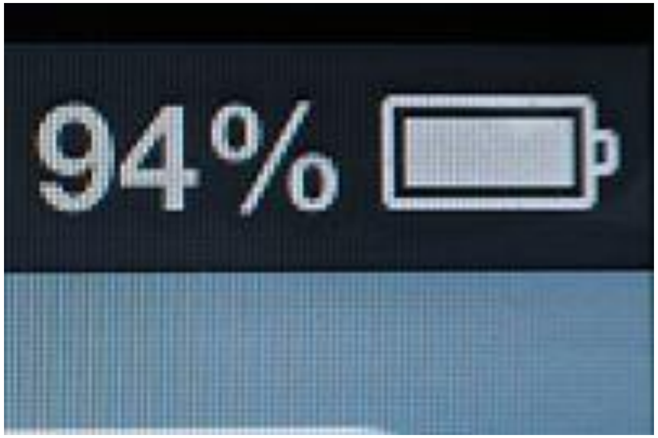



```
h = 2 × 0.4 × 0.000145444 = 0.116mm (at 0.4m)
h = 2 × 0.7 × 0.000145444 = 0.203mm (at 0.7m)
```

We can see individual pixels larger than about 0.116 to 0.203 mm

“Retina” Displays

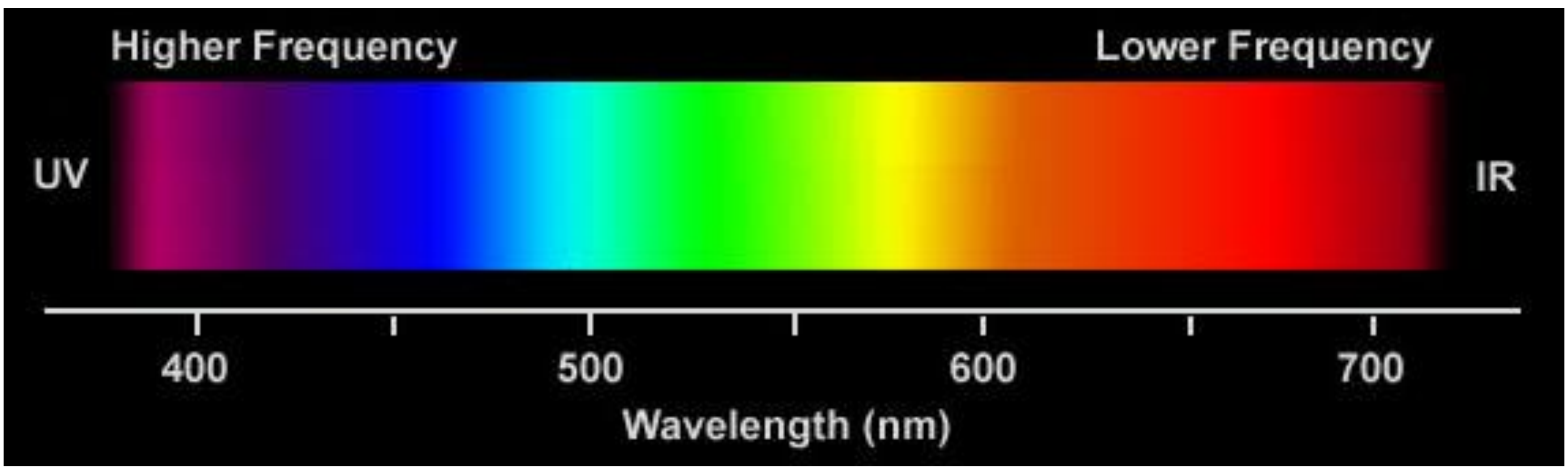
- iPhone X “Super Retina”
  - 2436 by 1125 px
  - 180 ppcm, pixel size 0.056mm



- 15 inch MacBook “Retina”
  - 2800 by 1800 px
  - 87ppcm, pixel size 0.12 mm

Visible Colour Spectrum

- Wavelength determines colour (in nanometres, nm)
  - Ultraviolet (UV)
  - Infrared (IR) (near IR used for input ~850nm)
- Combined wavelengths
  - example: orange is around 600 – 620 nm, but “orange light” can be brighter/darker when other wavelengths added

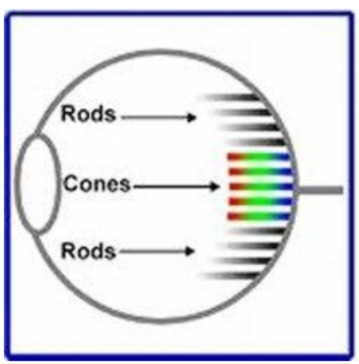


Interpreting the signal - colour

- Colour can improve user interfaces, inappropriate use of colour can be detrimental
- Seeing color depends on:
  - the nature of light
  - the interaction of light and matter
  - the physiology of human vision
- Process:
  - Light source strikes a coloured object
  - Object reflects light in another particular distribution of coloured wavelengths
  - Received by the photoreceptors of the human eye
  - Sent as a stimulus to the brain, causing us to perceive colour

Colour Perception

- Two different light sensors in human eye
  - **Cones** perceive colour (focus, ~7 million)
  - **Rods** distinguish light from dark (peripheral vision, ~120 million)
- Cones and rods not evenly distributed across the retina
  - spatial resolution of visual field drops significantly at edges



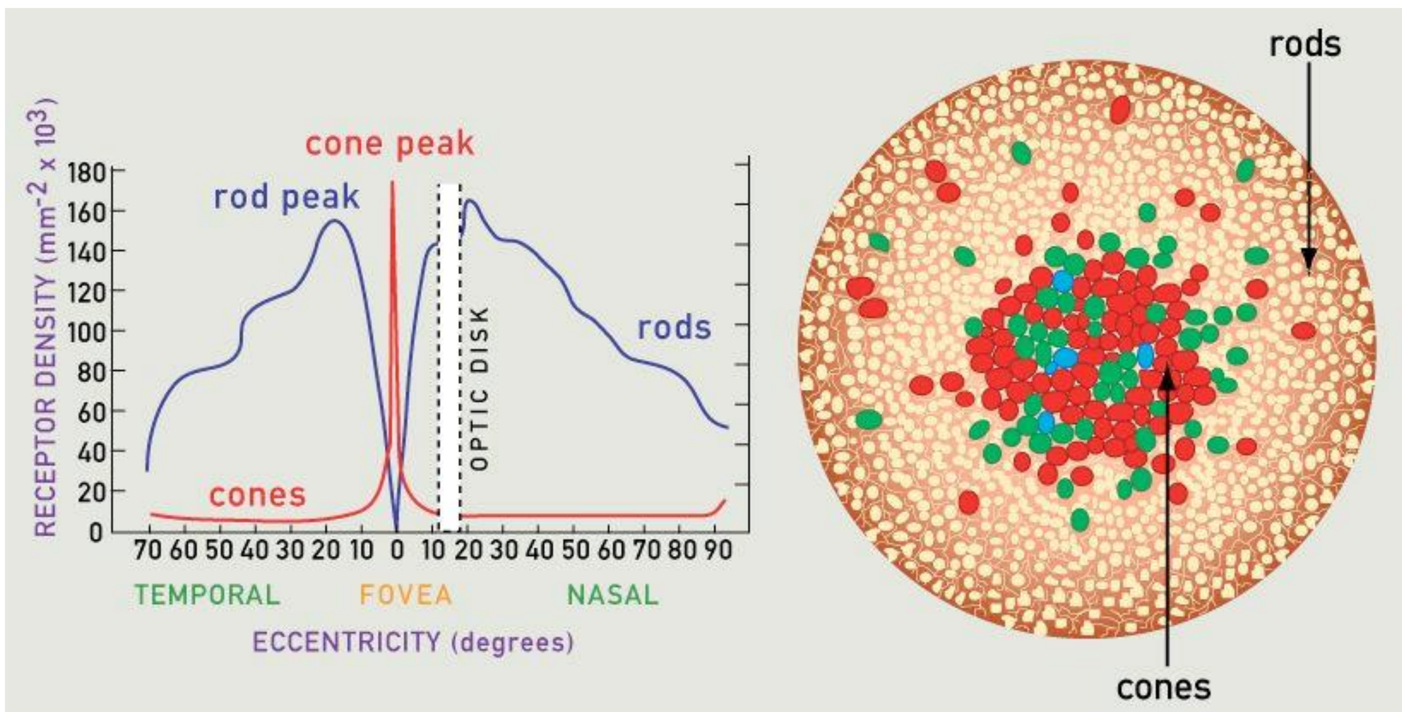
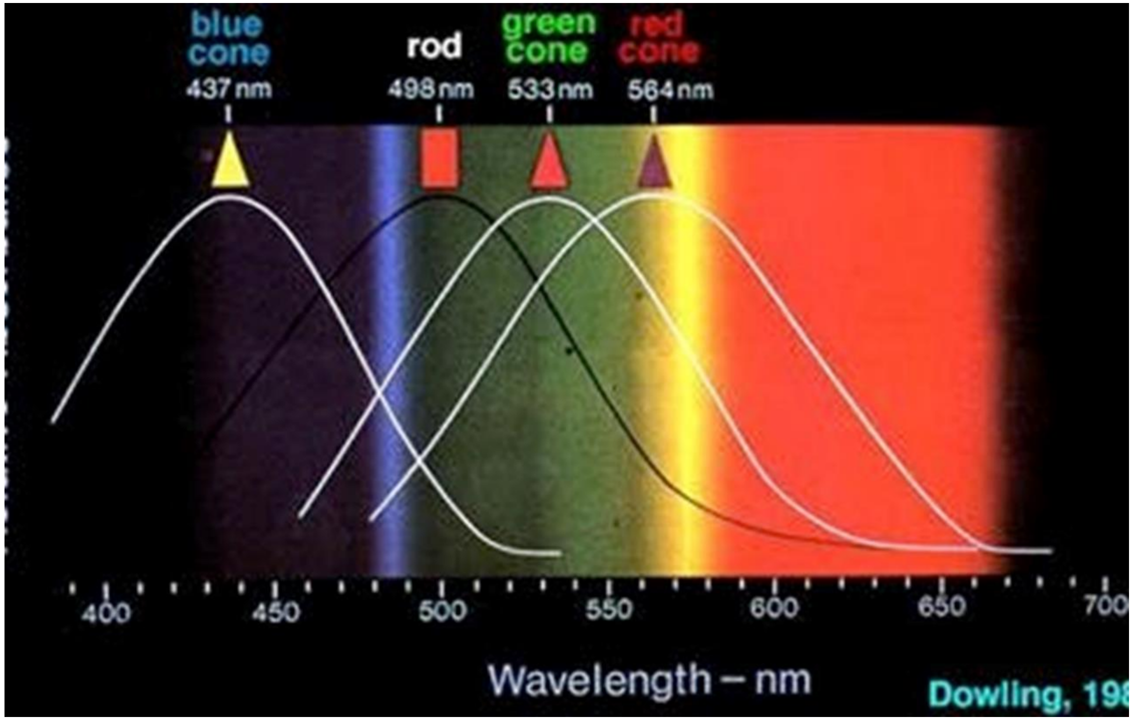


Image: www.webexhibits.org

3 Types of Cones means Trichromatic Vision



- Blue, green, and red cones
- Each is sensitive to a different band in the spectrum
- Few blue cones (approx: 64% red, 32% green, 4% blue)
- Harder to notice blues than reds

Humans, Birds, and Bees

Visible Light

UV Light

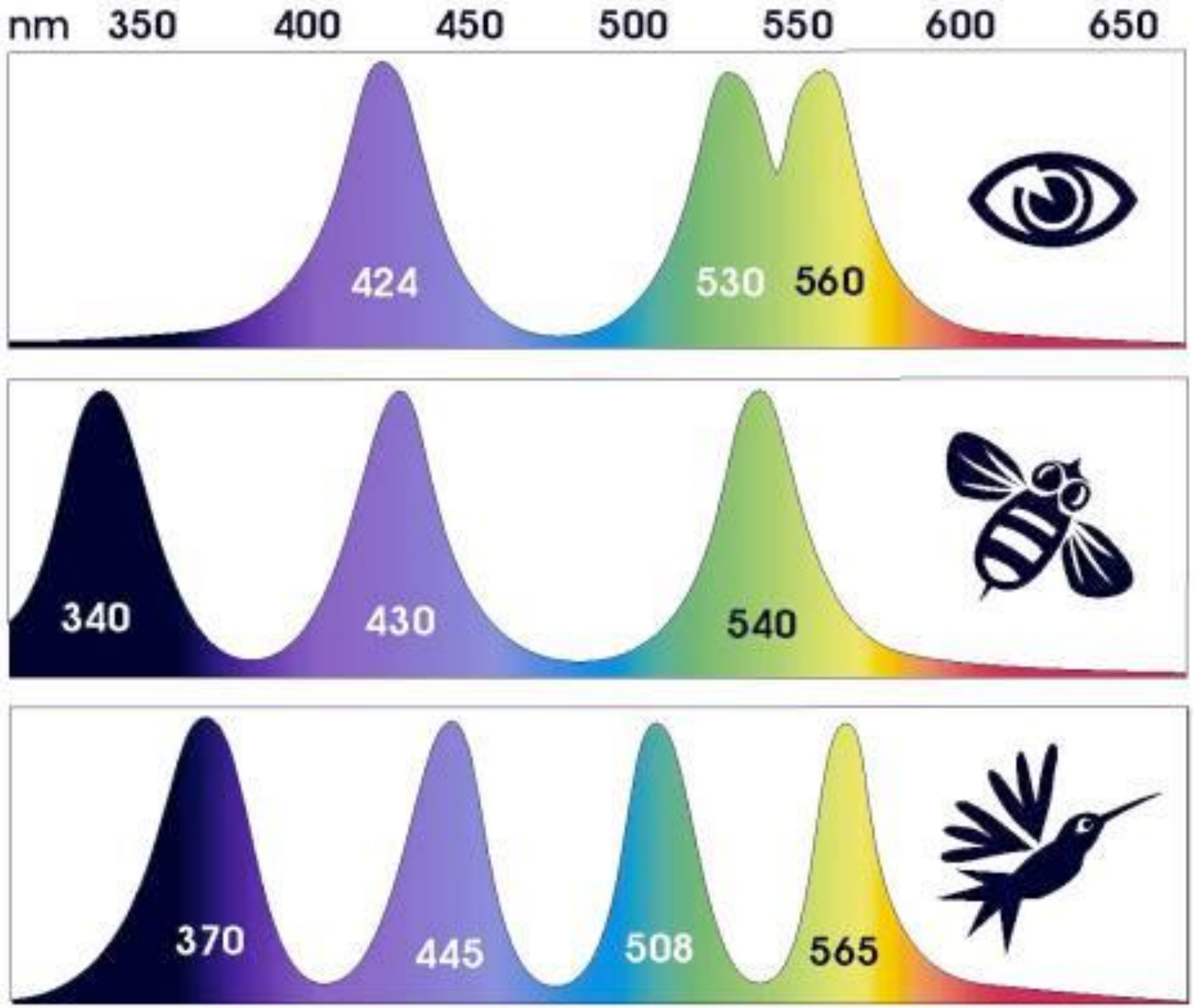
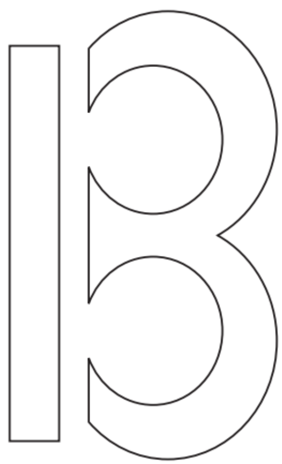


Image: http://fieldguidetohummingbirds.wordpress.com
http://www.naturfotograf.com/UV_flowers_list.html#top/

Capabilities and Limitations of Visual Processing

- The visual system compensates (to some degree) for:
  - movement
  - changes in luminance
- Context is used to resolve ambiguity



An ambiguous shape?

ABC

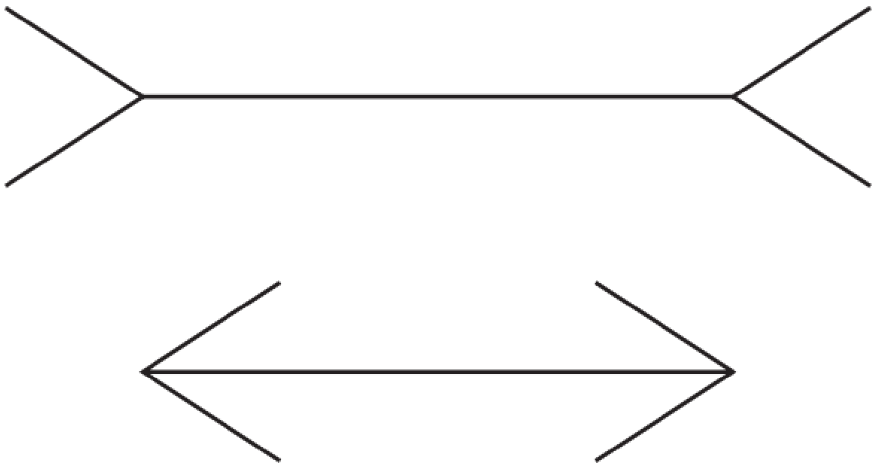
What color is the dress?

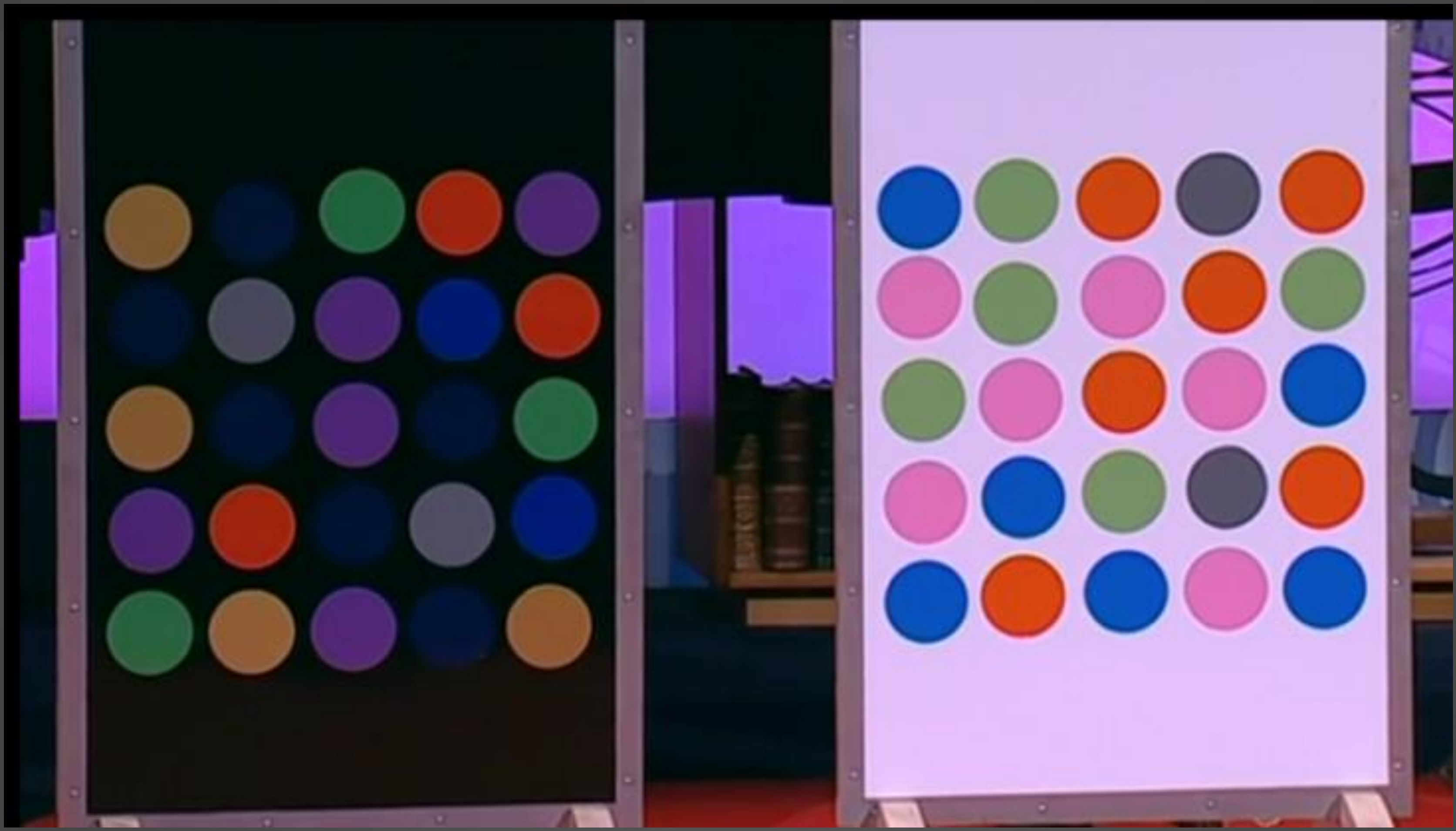
"your visual system is looking at this thing, and you're trying to discount the chromatic bias of the daylight axis. ... people either discount the blue side, in which case they end up seeing white and gold, or discount the gold side, in which case they end up with blue and black." - Jay Neitz, Professor, University of Washington

http://www.wired.com/2015/02/science-one-agrees-color-dress/
https://en.wikipedia.org/wiki/The_dress

The capabilities and limitations of visual processing

- Optical illusions sometimes occur due to over-compensation
  - movement
  - changes in luminance
- Most people say that the top line is longer than the bottom
- In fact, the two lines are the same length.
- This may be due to a false application of the law of size constancy: the top line appears like a concave edge, the bottom like a convex edge.



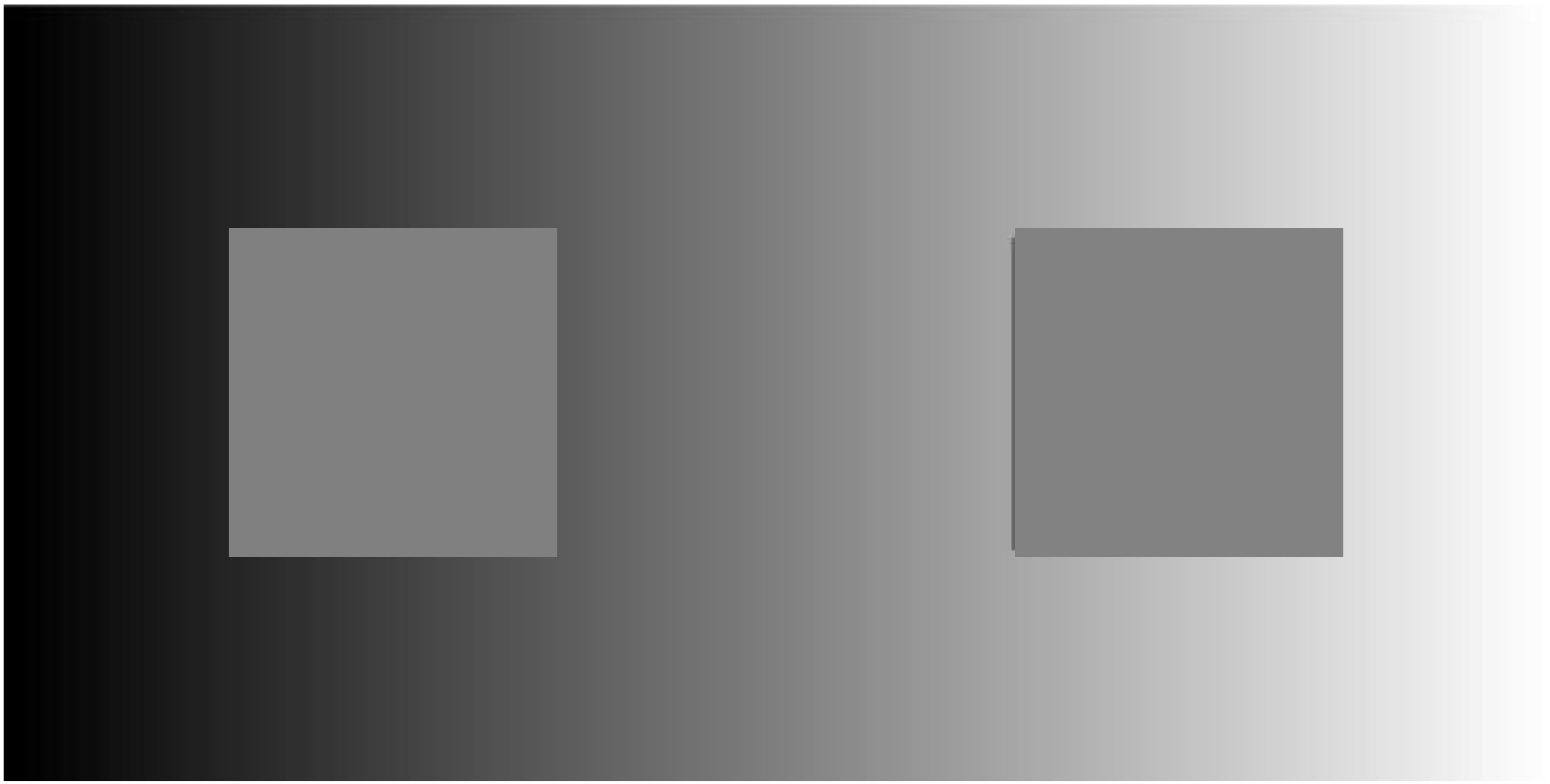


Beau Lotto, **Optical illusions show how we see (Ted 2009)**

- https://www.ted.com/talks/beau_lotto_optical_illusions_show_how_we_see#t-461256

Contrast, not Brightness

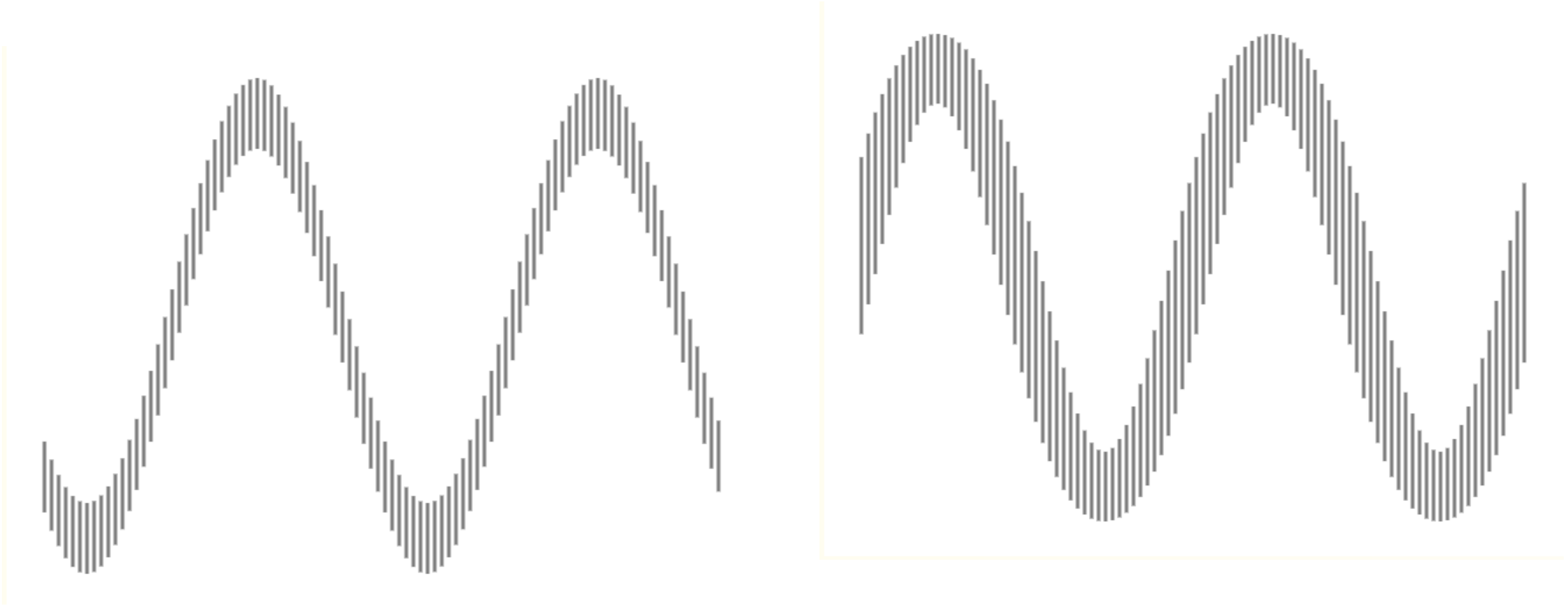
- We are more sensitive to differences in colour and brightness, than absolute brightness levels.



http://www.psy.ritsumei.ac.jp/~akitaoka/ECVP2005b.html

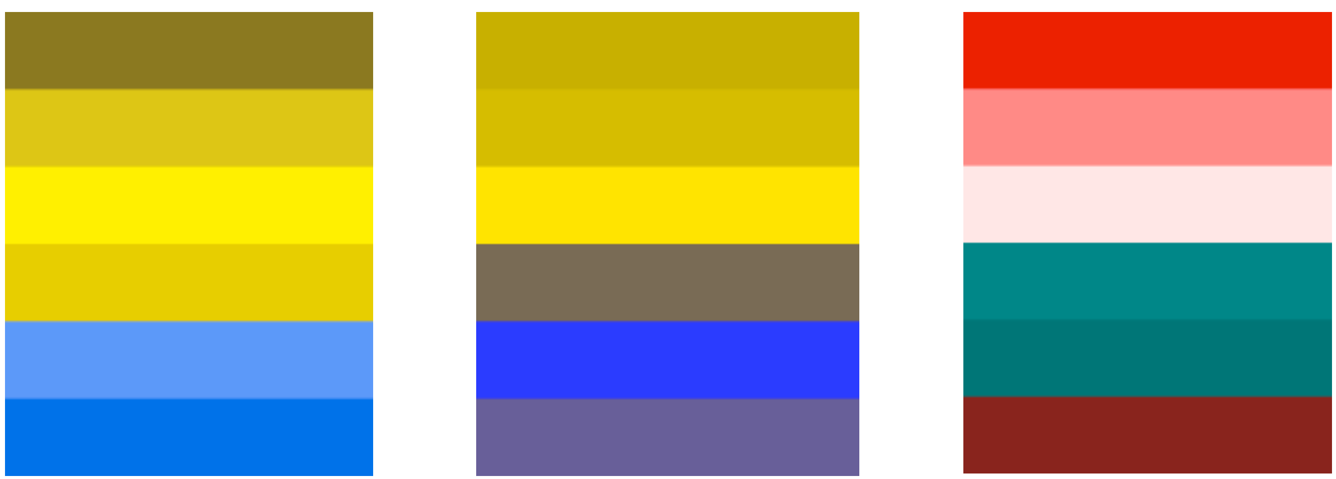
Visual illusions

- Visual Phenomena & Optical Illusions with explanations
- https://michaelbach.de/ot/index.html



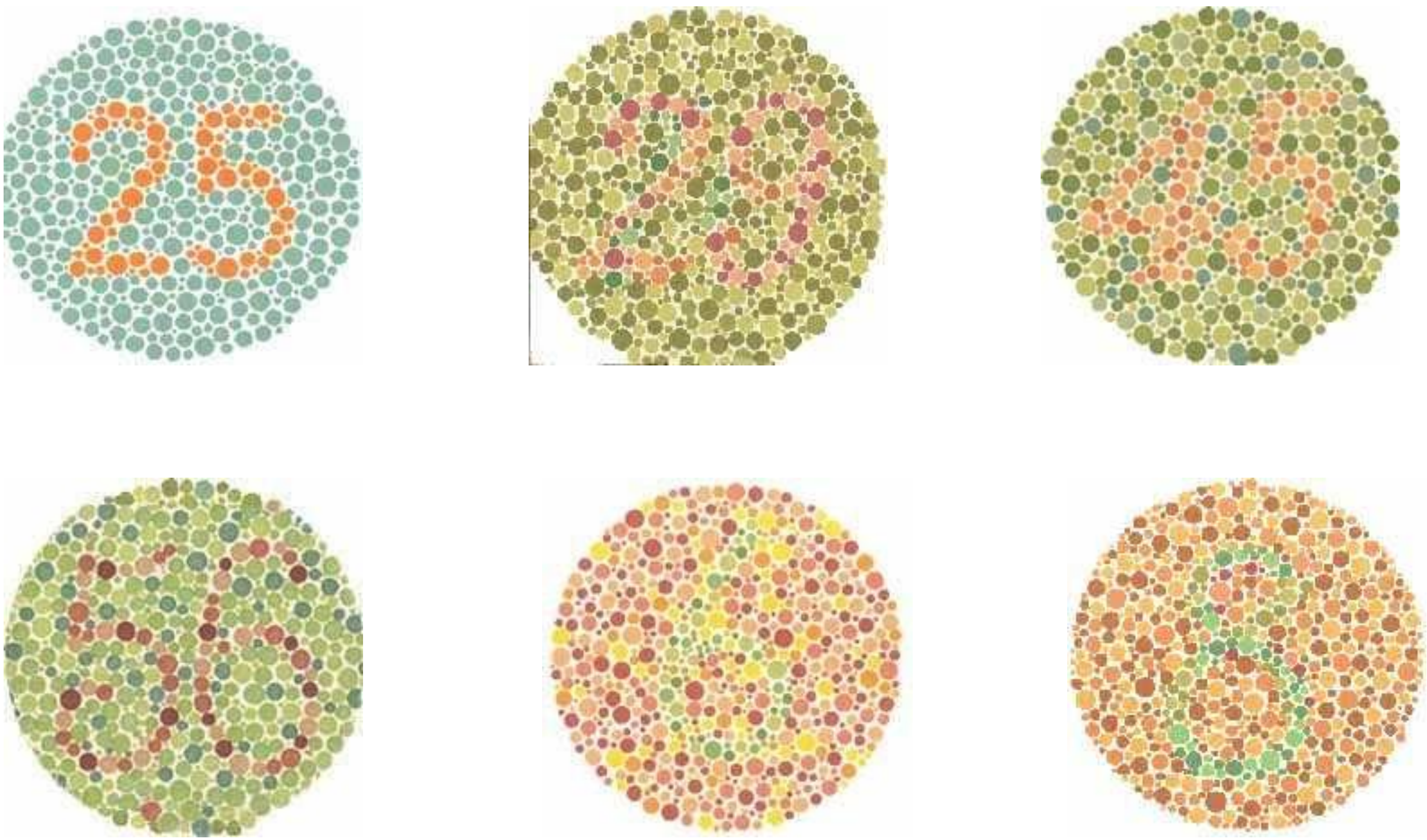
Colour Blindness

- Approximately 8% of human males, along with a rare sprinkling of females, have some form of colour blindness
- Most common is known as red/green colour blindness
- monochromacity: 2 or 3 types of cones missing
- dichromacy: 1 type of cone missing
  - Protanopia: missing red cones
  - Deuteranopia: missing green cones
  - Tritanopia: missing blue cones, (and blue sensitive rods) (rare)



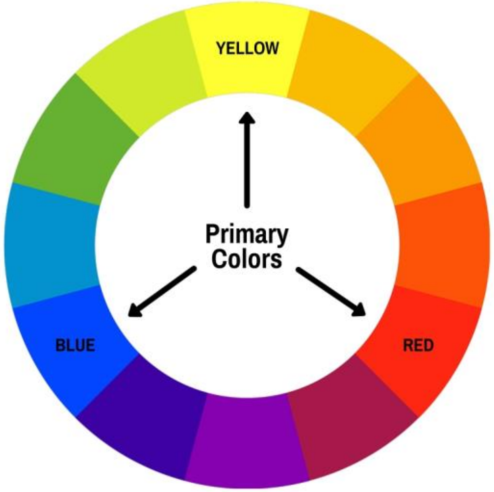
protanopia deuteranopia tritanopia

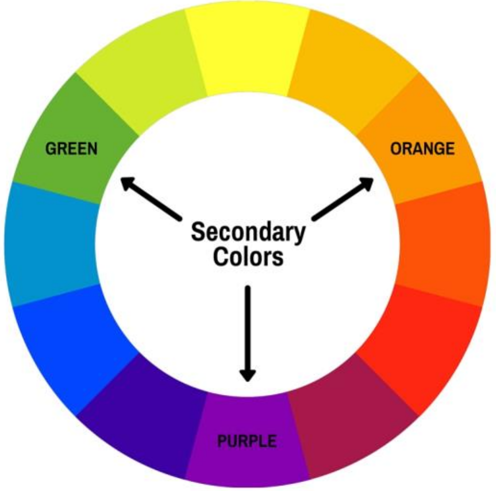
Ishihara Test for Colour Blindness

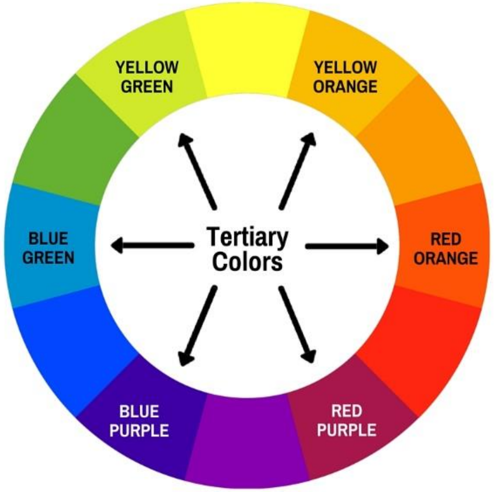


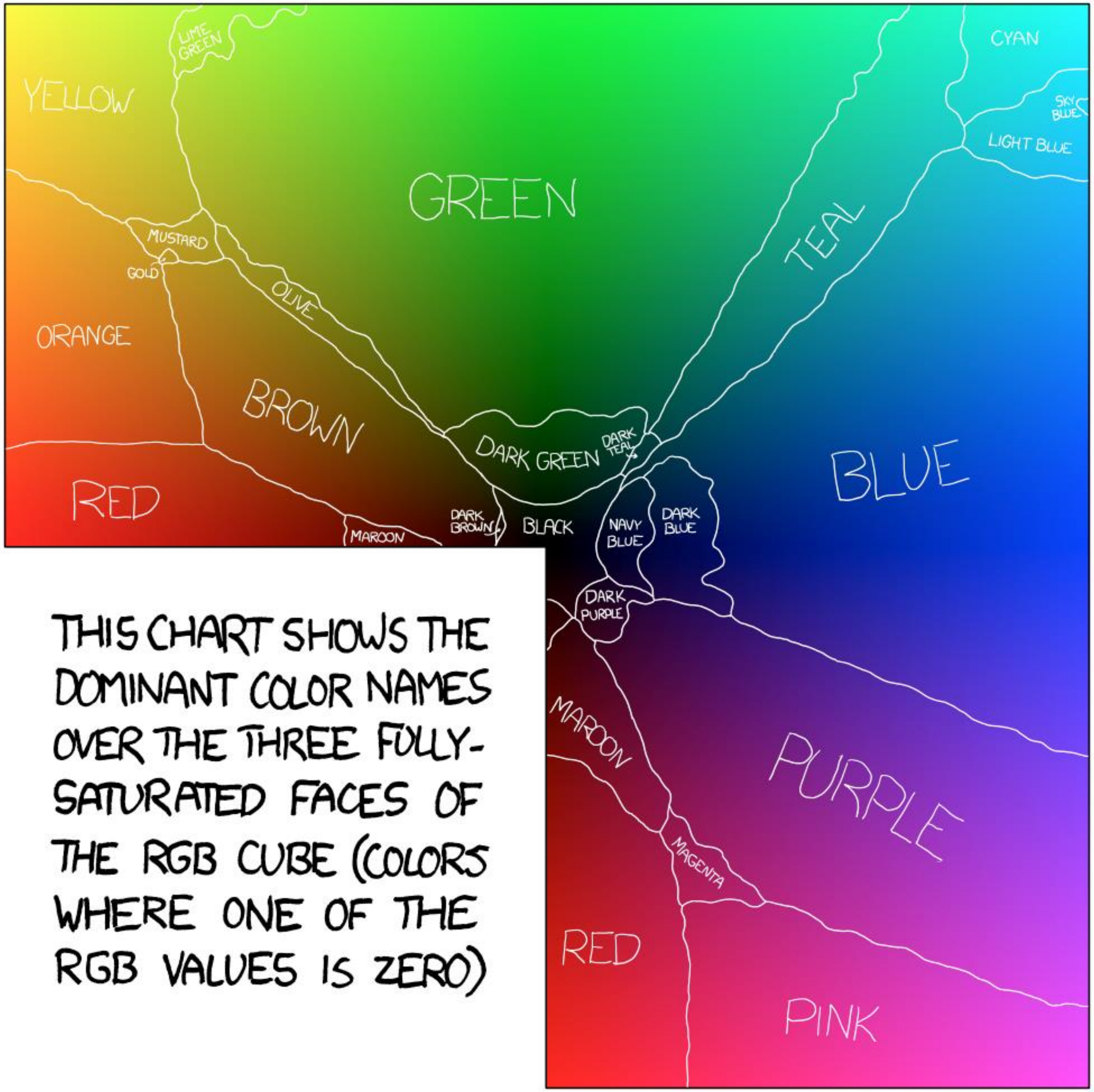
The Color Wheel

- The basic building block of color theory
- **Primary Colors**: Primary colors cannot be formed by mixing other colors. Typically red, yellow, and blue.
- **Secondary Colors:** Secondary colors are created by mixing two primary colors. Secondary colors are orange, green, and purple.
- **Tertiary Colors:** Tertiary colors are a combination of both a primary color and a secondary color. They often have two-word names, like "red-orange" or "blue-green"







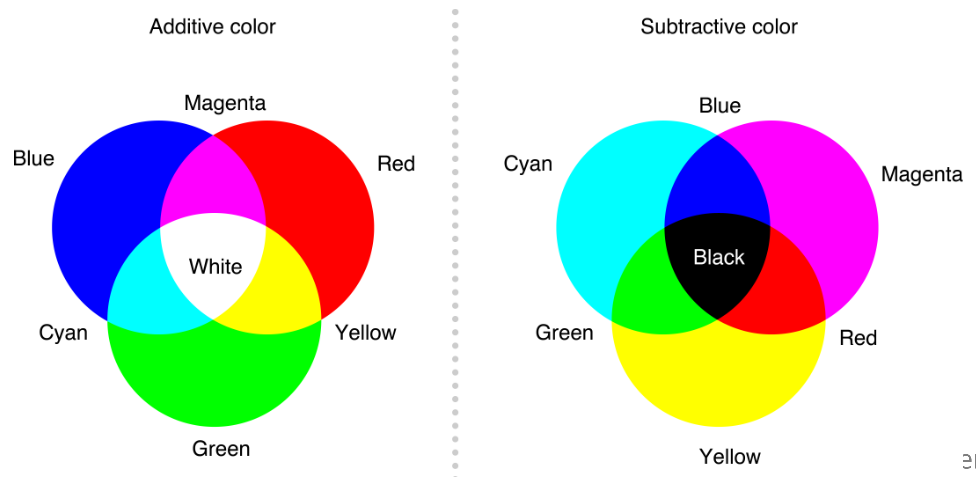


XKCD Colour Survey Results, https://blog.xkcd.com/2010/05/03/color-survey-results/

Colour Models

- **Additive**
  - coloured light is added to produce white
- TVs, computer monitors, and other electronics use additive color
- Every pixel starts as black, and take on colors that are expressed as percentage values of red, green, and blue (RGB)

- **Subtractive**
  - coloured light is absorbed to produce black
- Subtractive colors begin as white.
- CMY/CMYK – Photos, magazines, and any printed material



Colour Presentation Matters

- Our ability to discriminate colours depends on presentation
- Example: it's harder to tell two colours apart when
  - the colours are pale
  - the object is small or thin
  - the colour patches are far apart

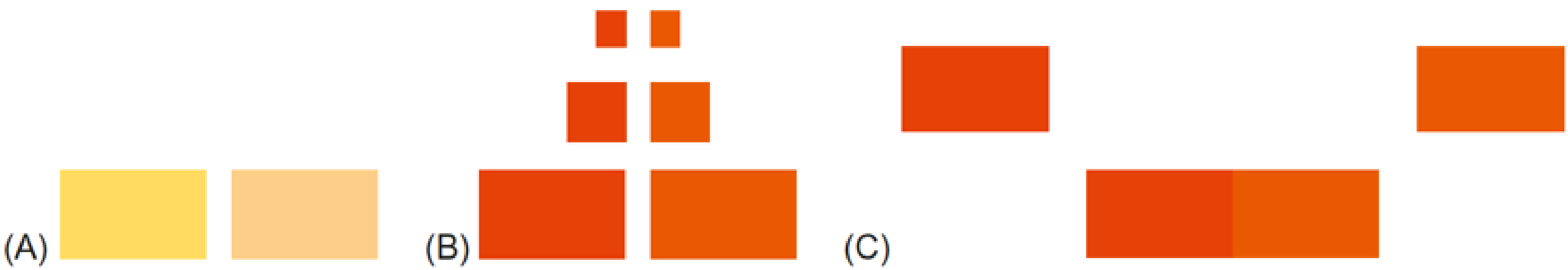
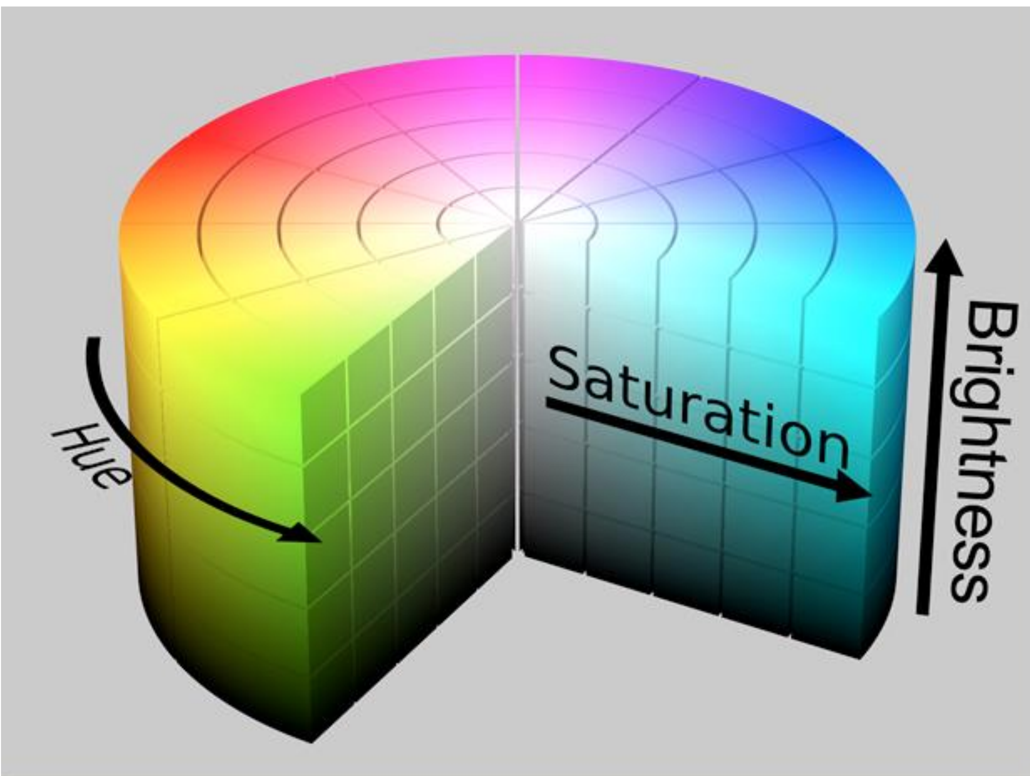


FIGURE 5.4

Factors affecting the ability to distinguish colors: (A) paleness, (B) size, (C) separation.

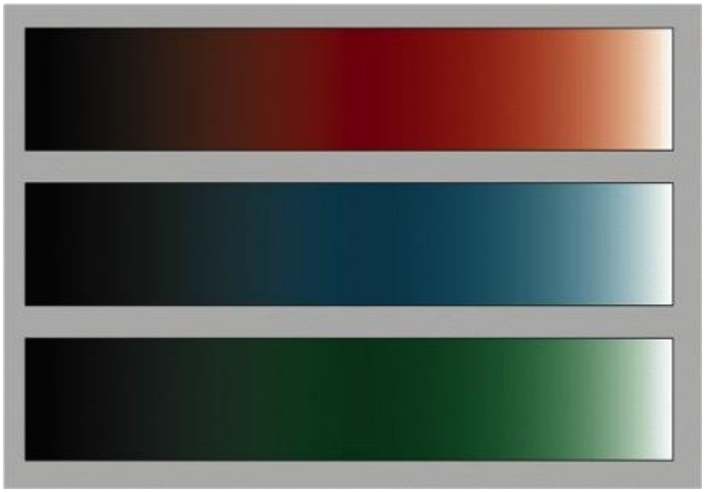
HSV/HSB Color model

- **Hue**: determines color (approximation of wavelength)
- **Saturation**: how much hue: e.g. red vs. pink vs. white
- **Value/Brightness**: how much light is reflected



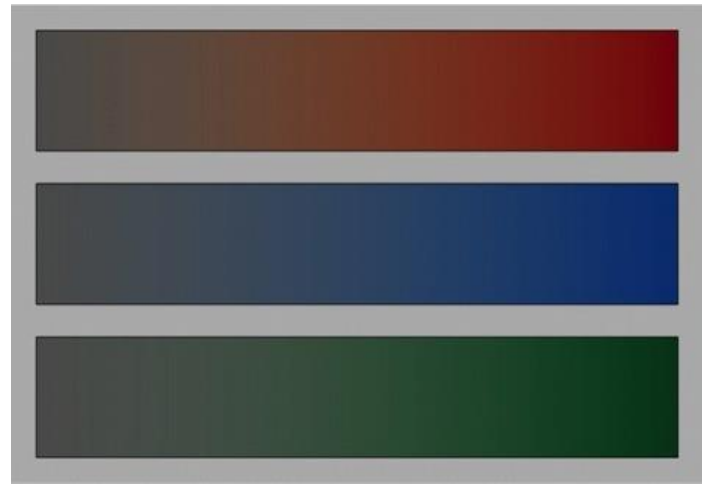
Value/Brightness vs. Saturation

- **Value/Brightness**
  - Reflecting less to more light



(black to white)
Fixed saturation,
changing value/brightness

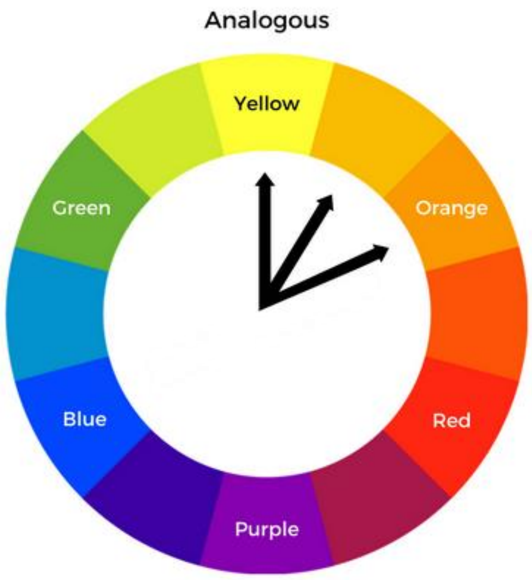
- **Saturation**
  - Containing less to more hue

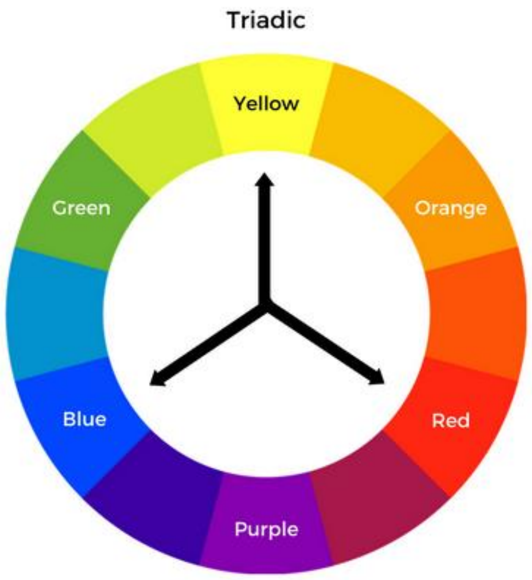


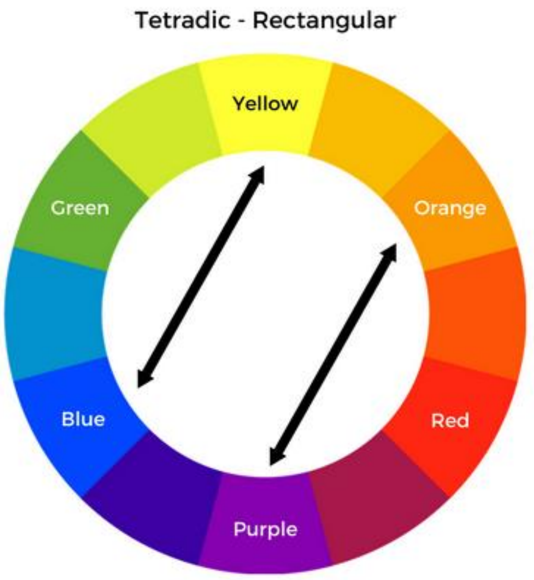
(gray to red, green, or blue)
Fixed value/brightness,
changing saturation

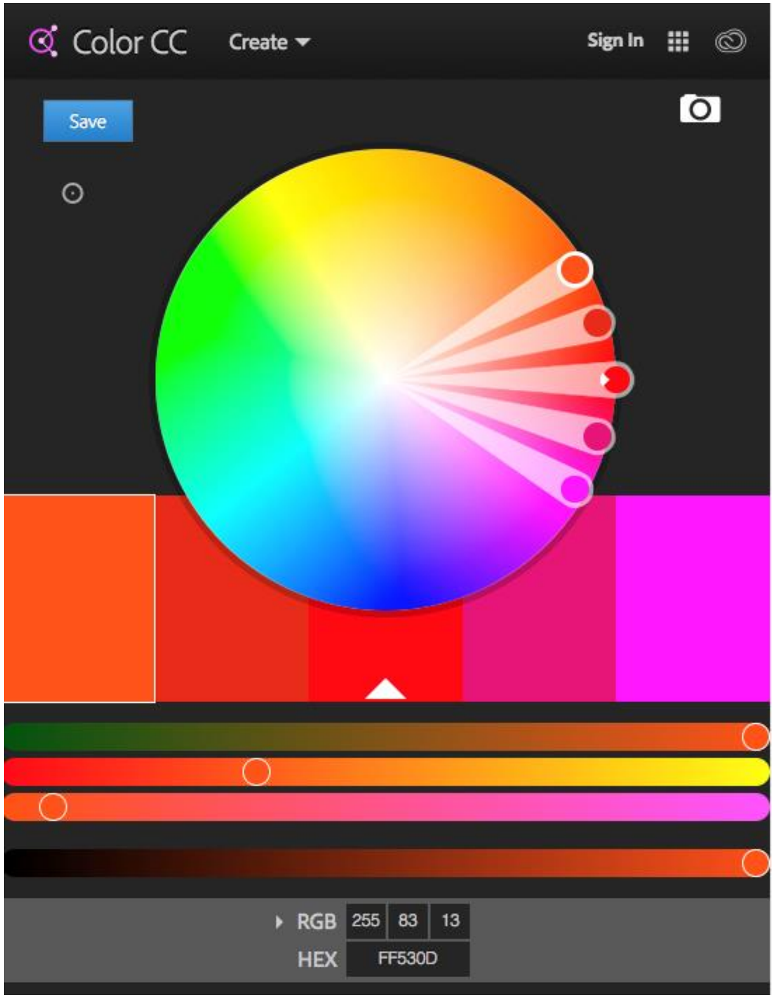
Colour Harmony

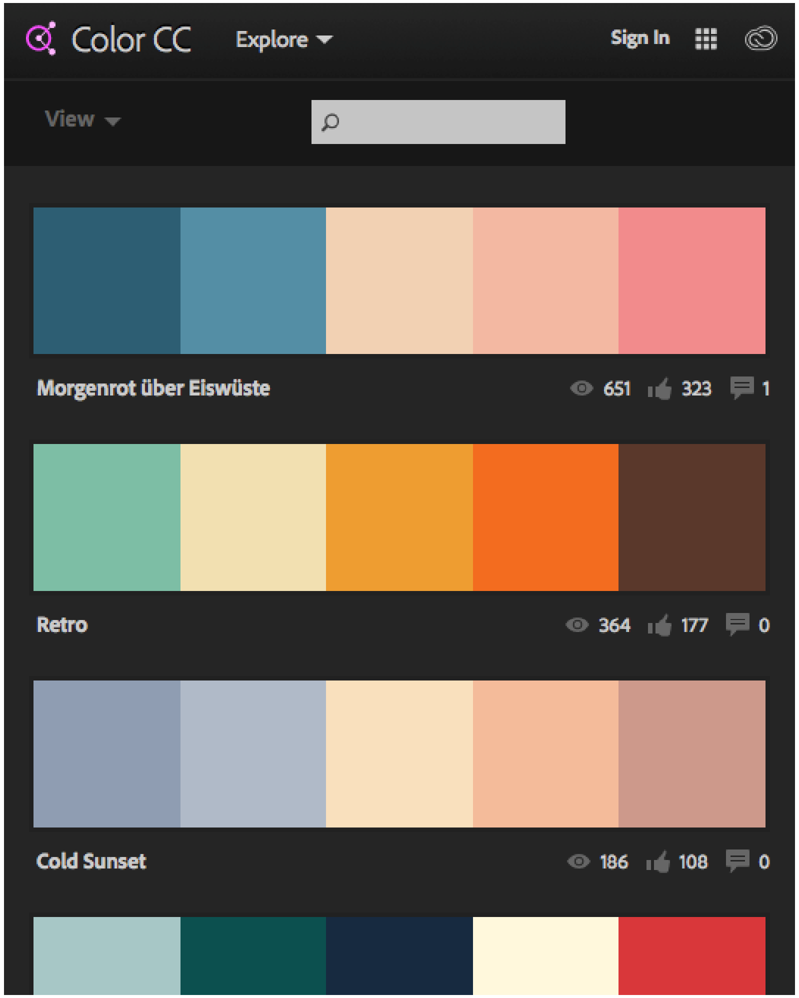
- **Analogous**: colors are located next to each other
  - When you are seeking a calm and unified look
- **Triadic**: three colors evenly spaced around the color wheel.
  - To create drama and contrast, bright designs
- **Tetradic**: four colors arranged in two complementary pairs.
  - Designs with color-coded elements, such as transit maps.









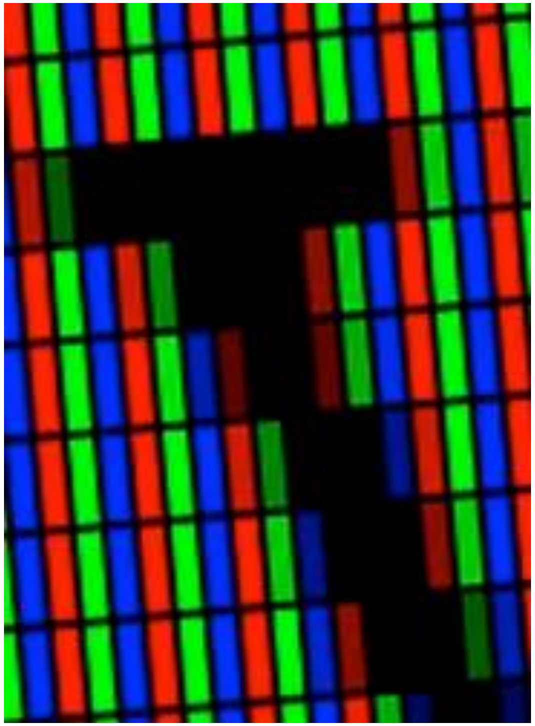


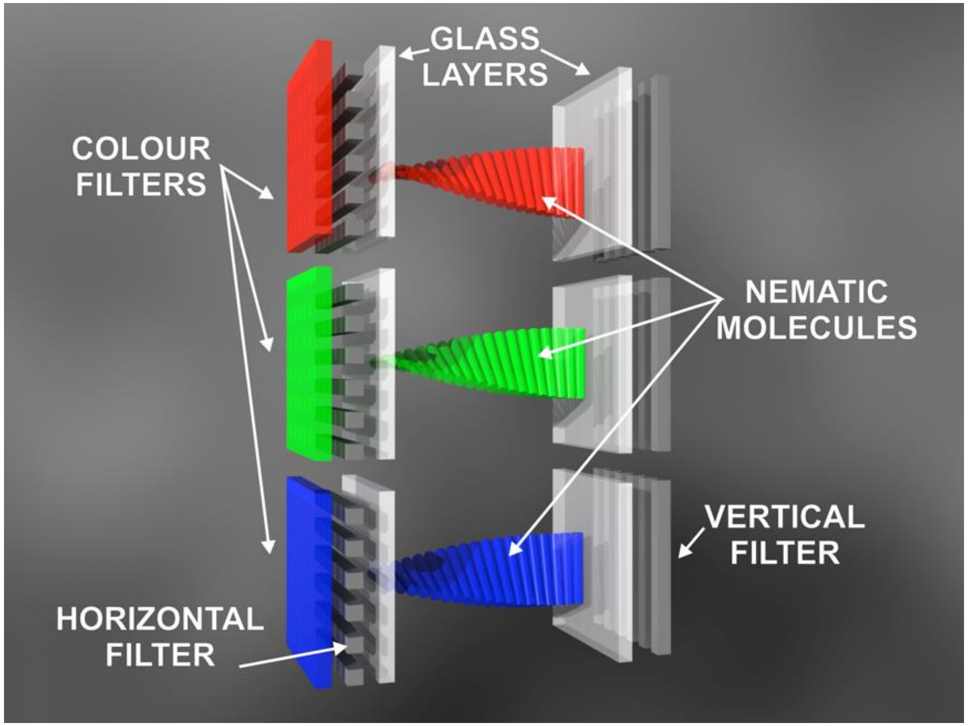
Adobe Colour Tool

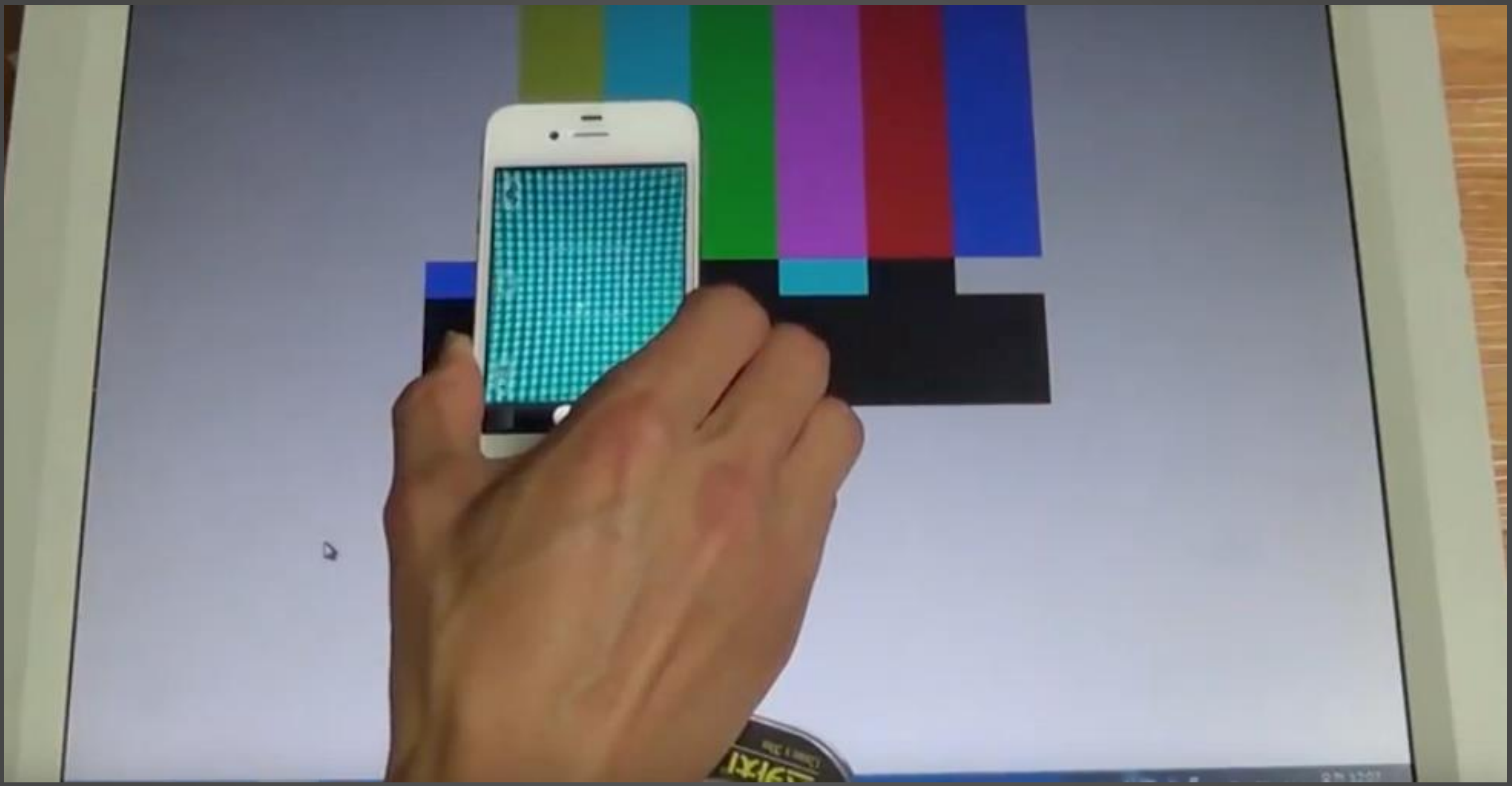
- https://color.adobe.com

Graphic Display Technology

- Common idea
  - Each pixel is actually 3 RGB sub-pixels: red, green and blue
  - Pack subpixels very close together so they seem to be co-located
  - Vary amounts of red, green, blue to excite cones in eyes







How to see your monitor subpixels

- https://youtu.be/_O66qHq1YS4

Use of Colour

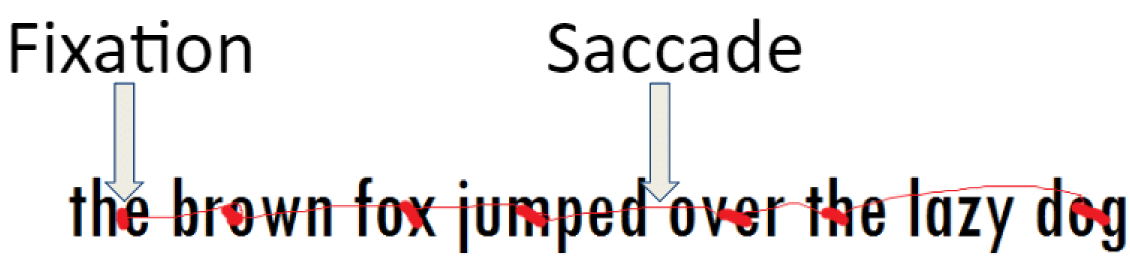
- Colour is an extremely valuable tool in design
- But when using colour, we also need to exercise caution
  - Cannot detect colour differences in the periphery
  - Certain colours more easily detectable (e.g., blue has low sensitivity)
  - Need to design for colour blindness
- People have emotional responses to color. E.g.,
  - Red regarded as a "hot" color that provokes aggression
  - Blues are cool and relaxing
  - White suggests purity
- There are also cultural differences
  - Examples?

Cultural Variables

	Europe & USA	China	Japan	Middle East
	Danger, Anger, Stop	Joy, Festive Occasions	Anger, Danger	Danger, Evil
	Caution, Cowardice	Honor, Royalty	Grace, Gaiety Nobility	Happiness, Prosperity
	Fertility, Safe, Sour, Go	Youth, Growth	Future, Youth, Energy	Fertility, Strength
	Purity, Virtue	Mourning, Humility	Death, Mourning	Purity, Mourning
	Masculinity, Calm,Authority	Strength, Power	Villainy	
	Death, Evil	Evil	Evil	Mystery, Evil

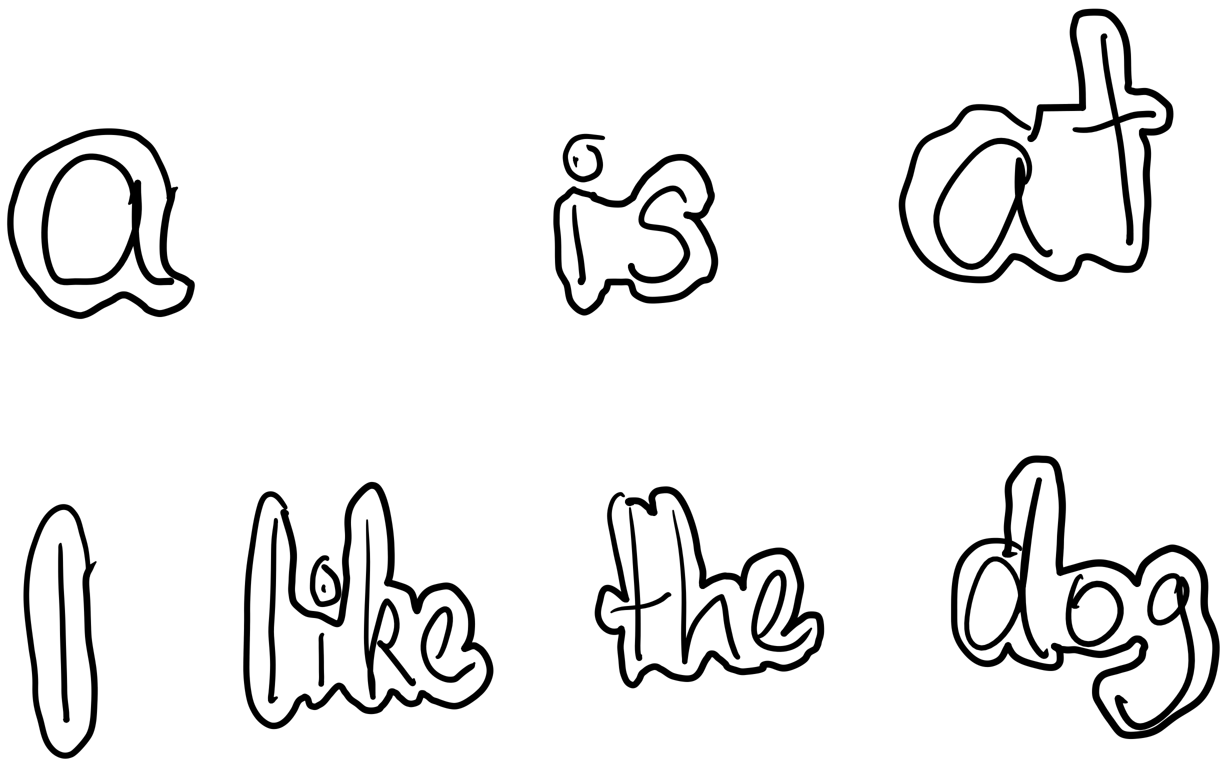
Reading

- Several stages:
  - visual pattern perceived
  - decoded using internal representation of language
  - interpreted using knowledge of syntax, semantics, pragmatics
- Reading involves saccades and fixations
  - A saccade is the rapid eye movement between fixations to move the eye-gaze from one point to another
  - A fixation is the point between two saccades, during which the eyes are relatively stationary and virtually all visual input occurs



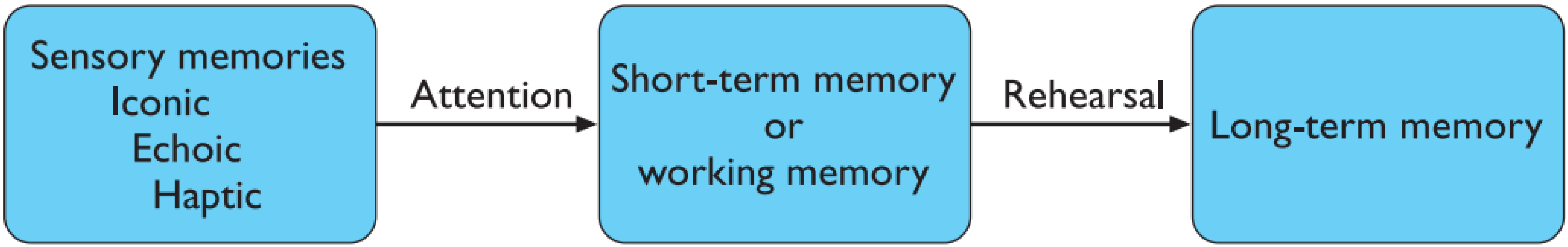
Reading

- Word shape is very important to recognition



Human Memory

- Memory is a core part of our model of the human as an information-processing system
- Involves encoding & recalling knowledge and acting appropriately
- We don't remember everything - involves filtering and processing
- Three types of memory or memory function:



Sensory memories

- The **sensory memories** act as buffers for stimuli received through the senses:
- One buffer for each channel:
  - *iconic memory* for visual stimuli
  - *echoic memory* for aural stimuli
  - *haptic memory* for touch
- Very brief, but veridical representation of what was perceived
  - Information constantly over-written/destroyed
  - Details decay quickly (~.5 sec)
  - Rehearsal prevents decay

Short-term Memory

- Short-term memory or working memory acts as a ‘scratch-pad’ for temporary recall of information.
- Can be accessed rapidly, however, it also decays rapidly
- Short-term memory also has a limited capacity.
  - **7 ± 2 chunks (Miller, 1956)**
- Chunk:
  - Any meaningful combination of items
- With good chunking, short-term memory can be extended
- E.g.
  - CAT DOG PIG CLOCK (n=4 chunks)

Long-term memory

- We store factual information, experiential knowledge, procedural rules of behavior – in fact, everything that we ‘know’
- Long-term memory is intended for the long-term storage of information.
- Information is placed there from working memory through rehearsal.
- It differs from short-term memory
  - it has a huge, if not unlimited, capacity
  - it has a relatively slow access time
  - forgetting occurs more slowly in long-term memory

Long-Term Memory

- Where is long-term memory important in HCI?
- Examples
  - Remembering procedures?
  - Websites visited?
  - Files?
- Reducing Memory load?
  - Users sometimes have to employ workarounds to reduce memory load
  - E.g., Externalizing information

Summary

- Temporal resolution
  - Critical Flicker Frequency
  - Temporal Resolution: Flicker into Motion
- Spatial Resolution
  - Visual Acuity
- Visible Colour Spectrum
  - Colour Perception
  - Trichromatic Vision

Question?